

Q1. A naive controller

$\tau = I\dot{\omega}$	(Newton)
$kE = I\dot{\omega}$	(response)
$k(\alpha U - S(\omega)) = I\dot{\omega}$	(instantiate)
$k(\alpha U - \omega + \omega_0) = I\dot{\omega}$	(naive sensor)
$\int \frac{d\omega}{\alpha U + \omega_0 - \omega} = \int \frac{k}{I} dt$	(separate)
$-\ln \alpha U + \omega_0 - \omega = \frac{k}{I}t + C$	(integrate)
$\omega(t) = \alpha U + \omega_0 - C'e^{-kt/I}$	(solve)

Taking the limit,

$$\omega_{ss} = \lim_{t \rightarrow \infty} \omega(t) = \boxed{\alpha U + \omega_0}.$$

Q2. The improved controller

Instantiate the control law:

$$I\dot{\omega} = k \left(\alpha U - \ln \frac{\omega}{\omega_0} \right).$$

At steady state $\dot{\omega} = 0$, so

$$k \left(\alpha U - \ln \frac{\omega_{ss}}{\omega_0} \right) = 0 \implies \ln \frac{\omega_{ss}}{\omega_0} = \alpha U \implies \omega_{ss} = \omega_0 e^{\alpha U}.$$

Convergence: rewrite the dynamics as

$$\dot{\omega} = \frac{k}{I} \left(\alpha U - \ln \frac{\omega}{\omega_0} \right).$$

If $\omega > \omega_{ss}$ then $\ln(\omega/\omega_0) > \alpha U$, so $\dot{\omega} < 0$ (decelerates). If $\omega < \omega_{ss}$ then $\ln(\omega/\omega_0) < \alpha U$, so $\dot{\omega} > 0$ (accelerates). In both cases the rotor is driven toward ω_{ss} , hence

$$\boxed{\lim_{t \rightarrow \infty} \omega(t) = \omega_0 e^{\alpha U}}.$$